

The Prevalence of Refractive Error and Outcomes of Standard One Primary School Children in the District of Hilir Perak Using the Rapid Assessment Visual Acuity Chart (RAVAC) Test as a Screening Tool

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INTRODUCTION

Uncorrected refractive errors represent a major cause of treatable childhood visual impairment, which significantly impacts students' classroom learning, development, and long-term educational potential. To enhance school health screenings, the Rapid Assessment Visual Acuity Chart (RAVAC) was developed. RAVAC replaces the traditional, time-consuming Snellen chart, offering a highly validated and significantly faster visual acuity screening tool tailored for school health teams.

MATERIALS AND METHODS

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Inclusion Criteria

- All selected Standard One primary school children in the district of Hilir Perak.
- Signed parental or guardian consent.

Exclusion Criteria

- Existing ophthalmology follow-up for known ocular conditions.
- Absent on the examination day.
- Refusal of visual assessment/eye examination.
- Inconsistent visual acuity results after 3 attempts.

STUDY FOUNDATION

Target Population: Standard One Students
The study focuses specifically on Standard 1 children enrolled in primary schools within the PKD Hilir Perak school team.

Study Design: Retrospective Cross-Sectional
This school-based study analyzes existing data from the year 2024 to determine the current state of refractive error prevalence.

Primary Tool: RAVAC Test
Visual acuity is measured using the Rapid Assessment Visual Acuity Chart (RAVAC) Test, a streamlined tool for efficient screening.

RESEARCH OBJECTIVES

Determine Prevalence of Refractive Error
The general objective is to calculate how common refractive errors are among Standard One children in the Hilir Perak District.

Identify Screening Outcomes
The specific objective is to evaluate and identify the clinical outcomes resulting from the initial RAVAC Test screenings.

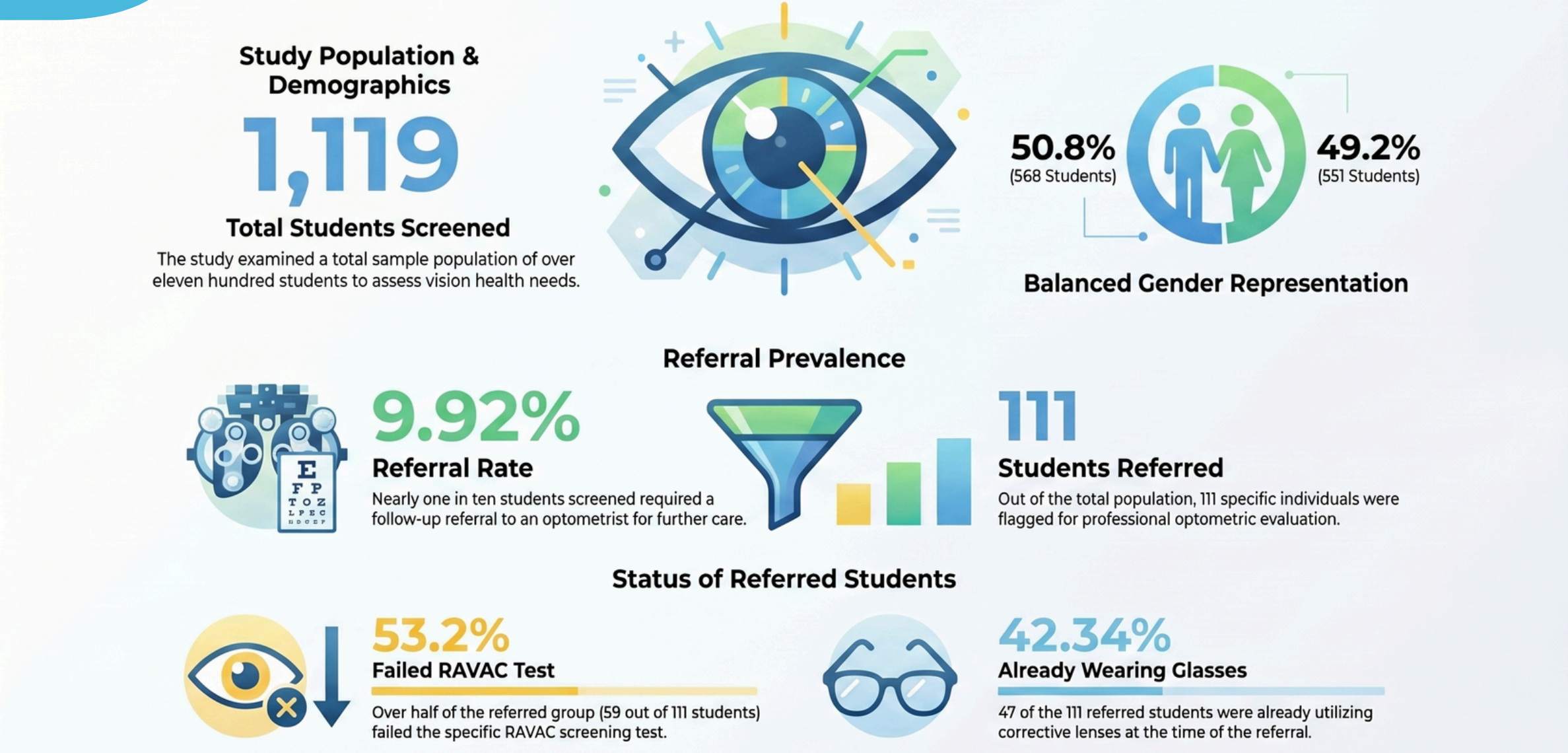
PROJECT TIMELINE

January 2024 – December 2024
This period marks the primary data collection phase for the RAVAC test results among the student population.

January 2025 – May 2025
A dedicated five-month window is established to track and analyze the time point frame of the RAVAC outcomes.

RESULTS AND OUTCOME

Student Vision Health: Study Screening Outcomes



CONCLUSION

RAVAC serves as a highly efficient and practical tool for identifying children who require further eye care in early education settings.

Nearly 1 in 10 screened students were referred for professional optometric evaluation.

Early identification and referral can support children's learning capabilities and aligns with the WHO Vision 2020 initiative to ensure every child's right to good vision.

ACKNOWLEDGEMENTS & REFERENCES

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Conflict of Interest: The authors declare no conflicts of interest regarding the conduct and publication of this screening study.

References:

1. World Health Organization (WHO). *Vision 2020: The Right to Sight - Global Initiative for the Elimination of Avoidable Blindness*. Action Plan 2006-2011.
2. Ministry of Health Malaysia (MOH). *Standard Operating Procedures for School Health Visual Screenings*.